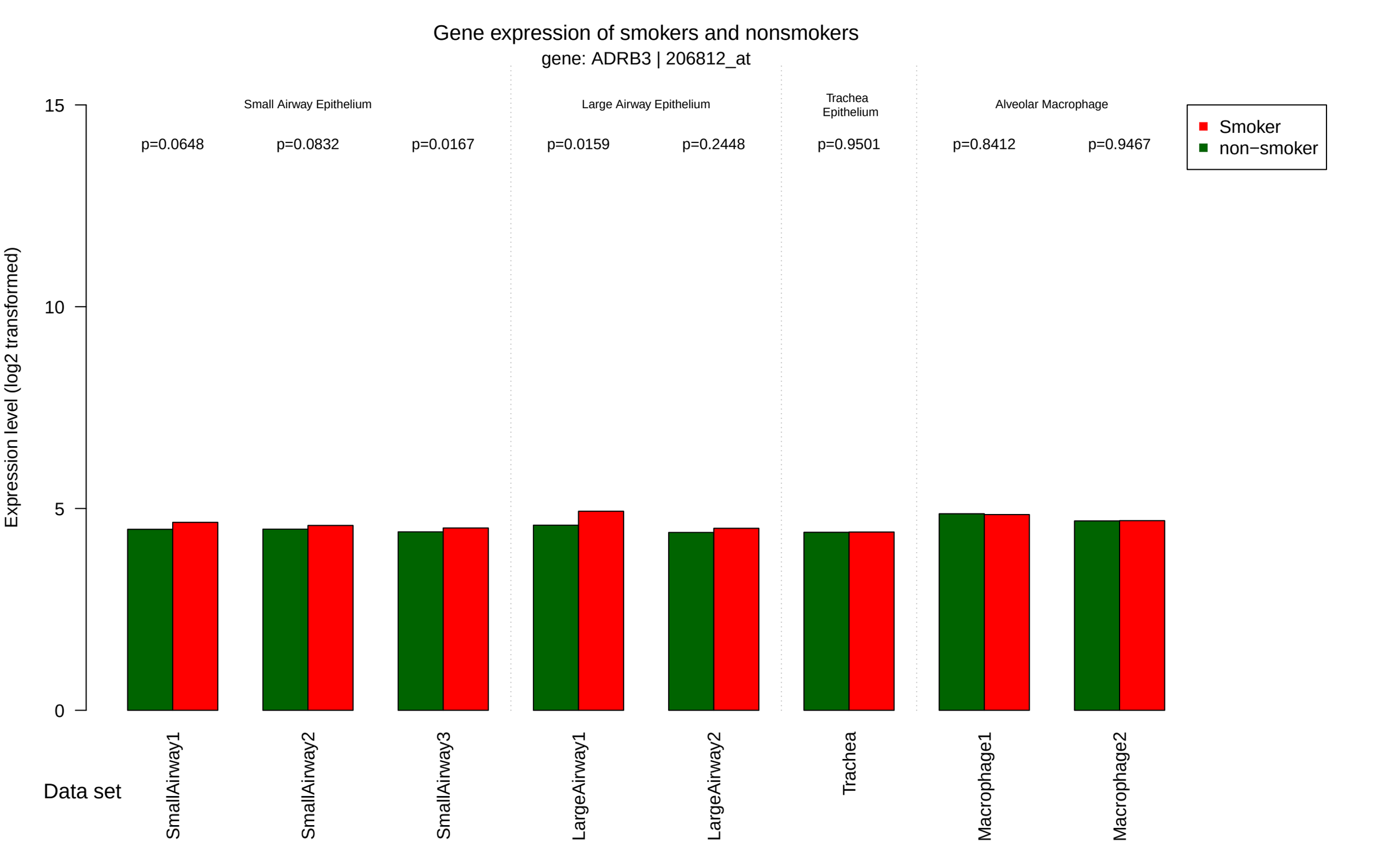
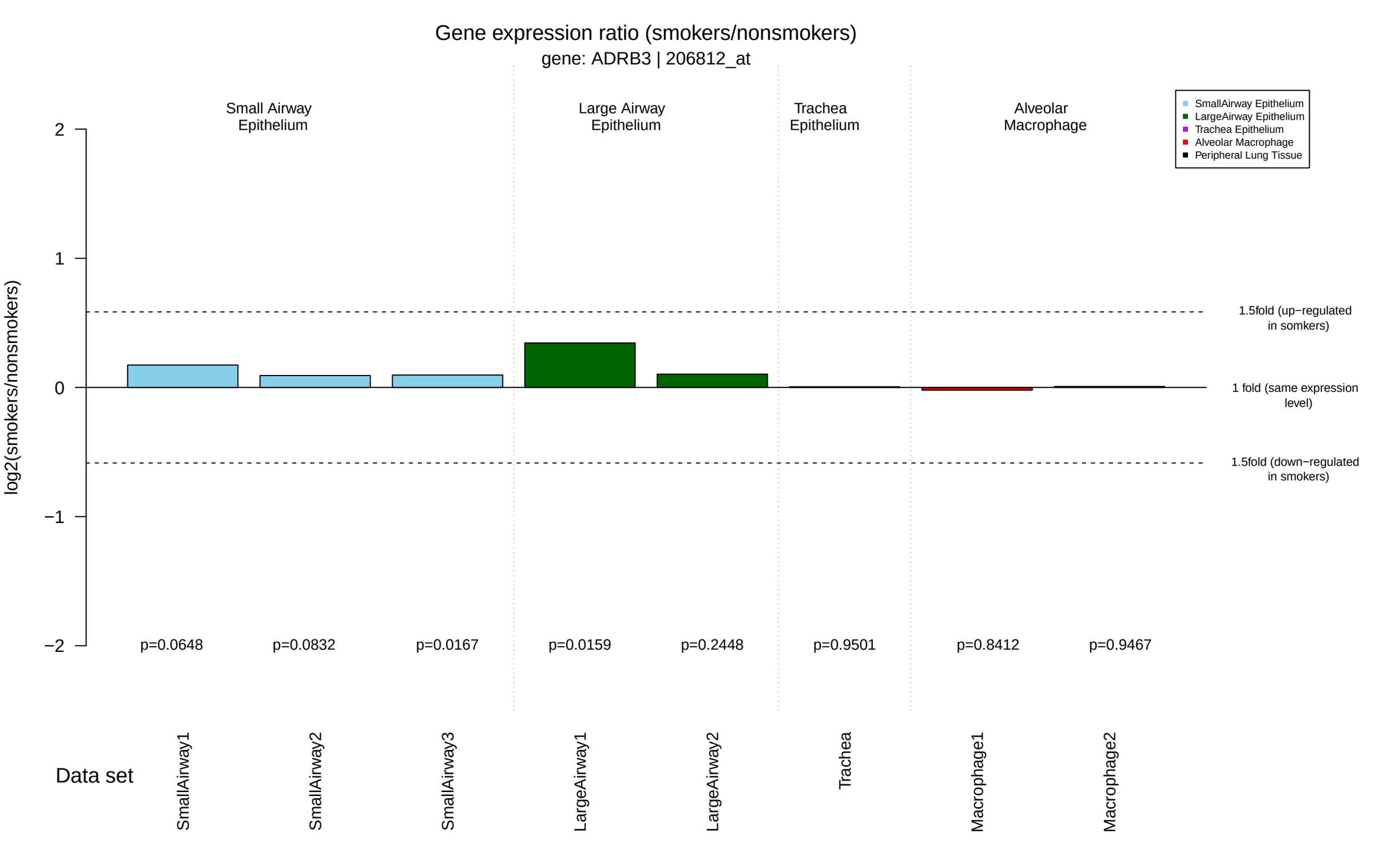
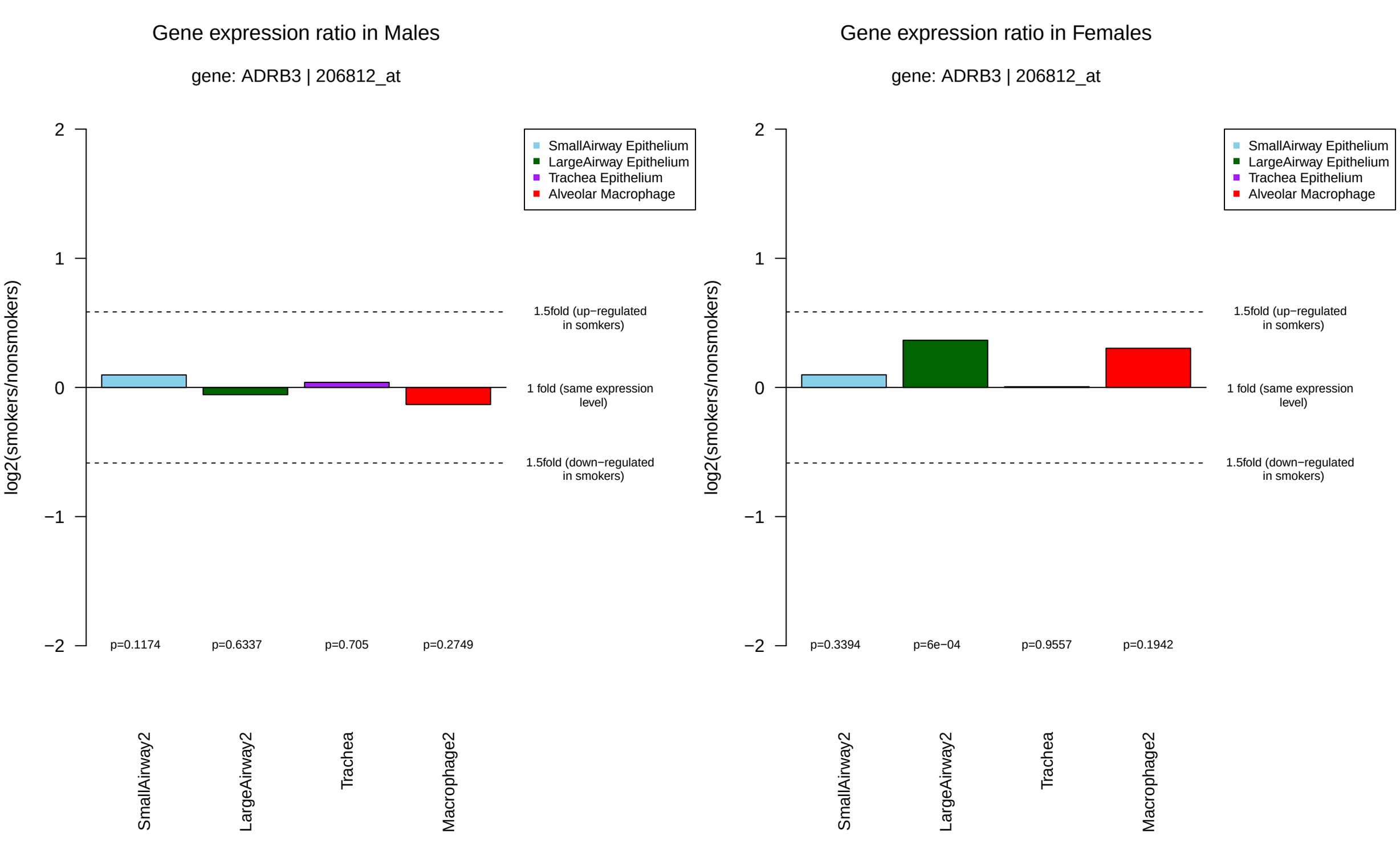
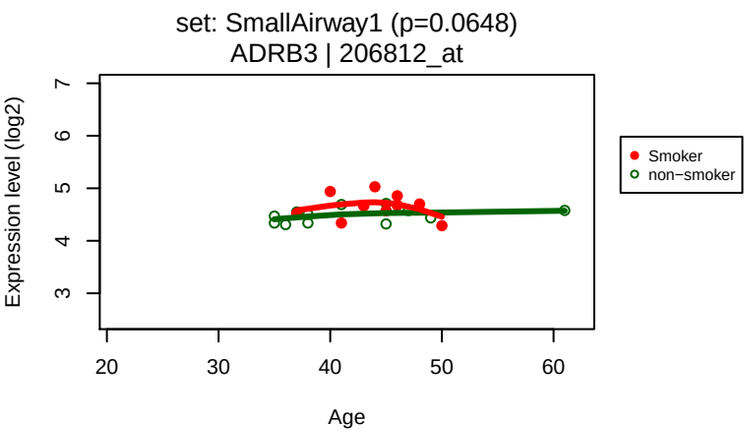
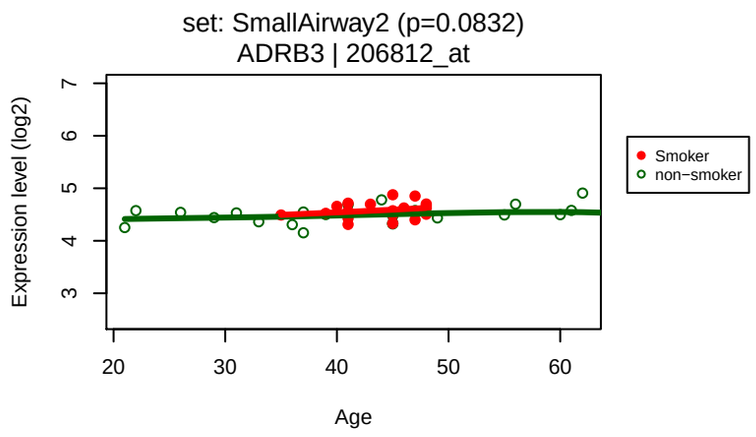
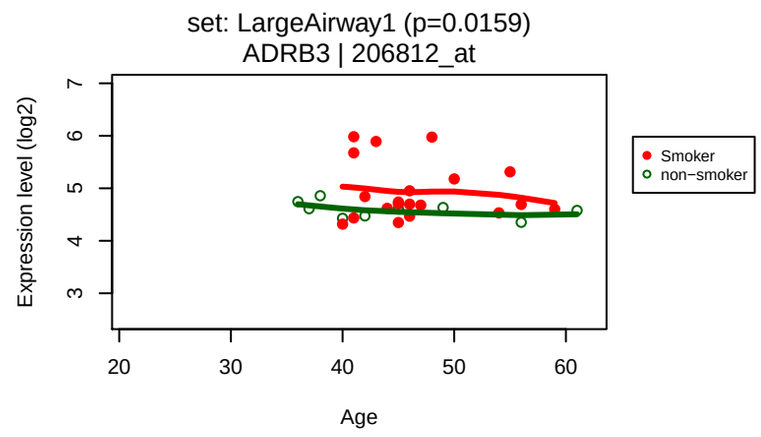

Gene expression of smokers and nonsmokers
gene: ADRB3 | 206812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0648
p=0.0832
p=0.0167
p=0.0159
p=0.2448
p=0.9501
p=0.8412
p=0.9467
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

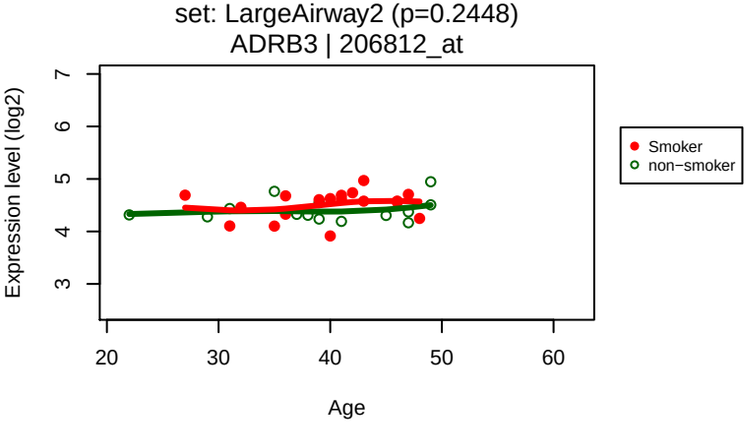

Gene expression ratio (smokers/nonsmokers)
gene: ADRB3 | 206812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0648
p=0.0832
p=0.0167
p=0.0159
p=0.2448
p=0.9501
p=0.8412
p=0.9467
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

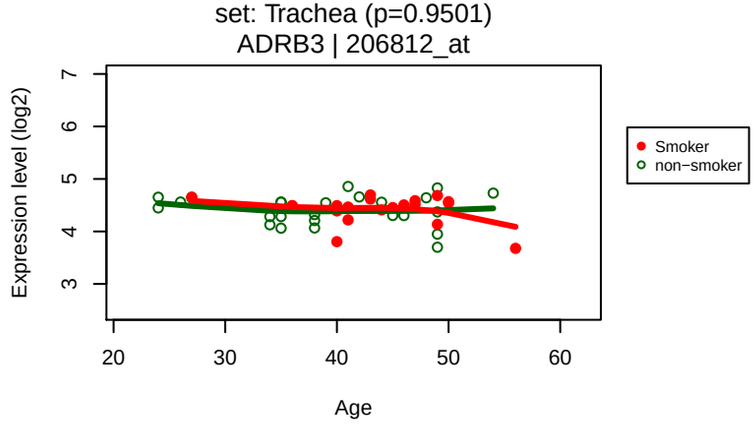

Gene expression ratio in Males
gene: ADRB3 | 206812_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1174
p=0.6337
p=0.705
p=0.2749
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADRB3 | 206812_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3394
p=6e−04
p=0.9557
p=0.1942
SmallAirway2
LargeAirway2
Trachea
Macrophage2

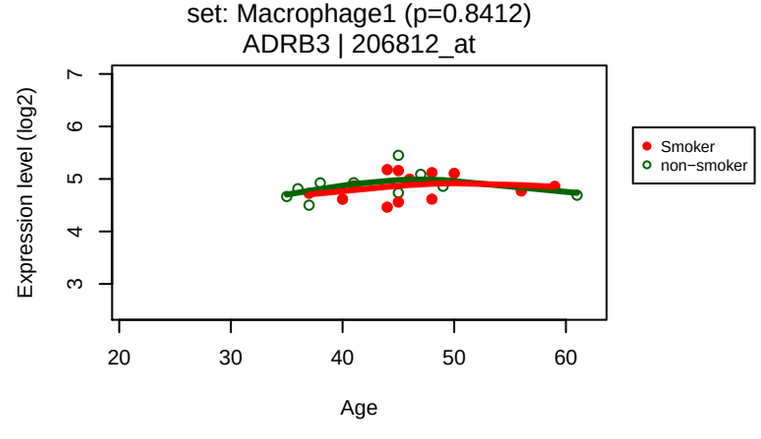

set: SmallAirway1 (p=0.0648)
ADRB3 | 206812_at
Expression level (log2)
Age
Smoker
non−smoker

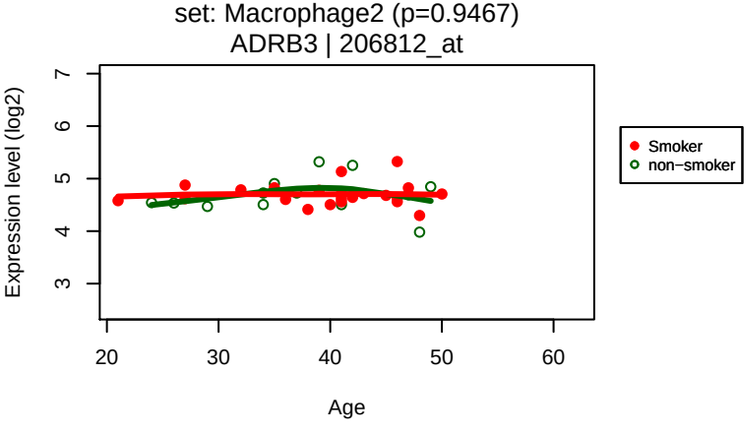

set: SmallAirway2 (p=0.0832)
ADRB3 | 206812_at
Expression level (log2)
Age
Smoker
non−smoker

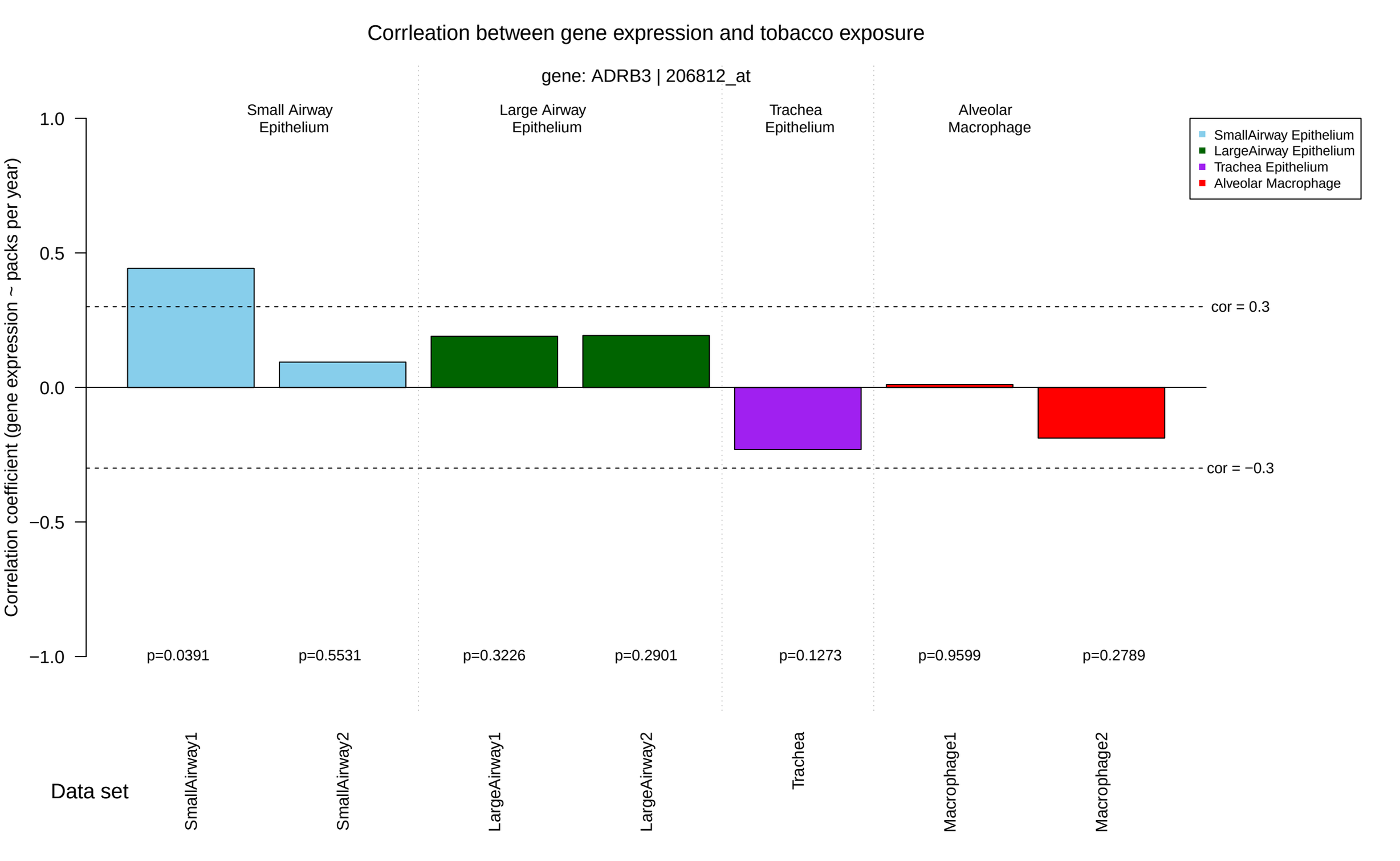

set: LargeAirway1 (p=0.0159)
ADRB3 | 206812_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2448)
ADRB3 | 206812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9501)
ADRB3 | 206812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8412)
ADRB3 | 206812_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9467)
ADRB3 | 206812_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADRB3 | 206812_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0391
p=0.5531
p=0.3226
p=0.2901
p=0.1273
p=0.9599
p=0.2789
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2